

Weekly Progress Report - Week 5

Improve performance of existing online trackers for small object detection [UAV Videos]

Group: The Convolutioners

{Aagam Sheth, Mahima Parekh, Khevan Thanki, Riddhi Bhargava}

Reporting Period: March 23 – March 28, 2026

Submission Date: March 28, 2026

Outline of Performed Tasks:

- Conducted a 12-configuration structured grid sweep over Siamese network hyperparameters
- Identified the best backbone architecture and training configuration from the sweep results
- Performed a threshold sweep to analyse the precision–recall trade-off
- Evaluated the fully optimised Siamese network on the held-out validation crop set
- Performed 12-trial Optuna Bayesian Optimisation in the Focal-BCE training regime
- Compared baseline and enhanced Focal-BCE loss variants and evaluated their effect on false positive reduction
- Compiled the best hyperparameter configuration for subsequent tracker integration

Grid Sweep: Hyperparameter Optimisation

A structured 12-configuration grid sweep was conducted to systematically identify the optimal Siamese network configuration. The search space covered the following hyperparameters:

- **Backbone:** ResNet18, ResNet34, MobileNetV3-Small
- **Embedding dimension d :** 64, 128, 256
- **Learning rate:** 10^{-4} , 3×10^{-4}
- **Contrastive margin m :** 0.8, 1.0, 1.2
- **Dropout:** 0.0, 0.2
- **Backbone freezing and L2 normalisation:** enabled / disabled

Optimised Training and Validation Performance

Using the best configuration identified from the grid sweep, the Siamese network was trained for 5 full epochs to assess performance under extended training. The aim of this stage was to observe how loss, accuracy, precision, recall, and F1 evolve as the model is given more time to refine its embedding space. In addition to per-epoch metrics, embedding distance statistics were tracked throughout training to confirm that same-object pairs are being pulled closer together while different-object pairs are being pushed further apart in the feature space, which is the expected behaviour under contrastive optimisation.

Threshold Sweep and Precision–Recall Analysis

A threshold sweep was conducted on the held-out validation set to examine how the choice of matching threshold τ affects the precision–recall trade-off of the Siamese Re-ID module. Since the tracker uses a fixed threshold to decide whether two embeddings correspond to the same identity, selecting an appropriate τ is critical: a threshold that is too low will miss valid matches and hurt recall, while a threshold that is too high will accept spurious matches and hurt precision. Six values of τ were evaluated to map out this trade-off and identify the value that achieves the best balance for tracker integration, prioritising the F1 score as the selection criterion. Any generalisation gap observed between training and validation performance is also noted and its likely causes discussed.

Optuna Bayesian Optimisation

Following the grid sweep, a 12-trial Optuna study using the Tree-structured Parzen Estimator (TPE) sampler was conducted as a complementary hyperparameter search in the Focal-BCE training regime. Unlike the grid sweep, which used contrastive loss and explored a fixed set of discrete configurations, the Optuna study samples hyperparameters intelligently from a continuous search space, focusing future trials on regions that have shown promise in earlier ones. The search space covered embedding dimension d , dropout rate, learning rate, negative pair weight w_{neg} , and toggle controls for L2 normalisation and backbone freezing. The objective metric was validation F1 after 3 epochs. This study serves to validate whether the design choices identified in the grid sweep are robust across a different loss formulation and a broader, continuous parameter space.

Alternative Focal-BCE Loss Comparison

As an alternative to the standard contrastive loss, a Focal Binary Cross-Entropy (Focal-BCE) formulation was explored to assess whether a classification-based training objective can offer complementary benefits. Two variants were compared:

$$L_{\text{FBCE}} = -[w_y \cdot y(1-p)^\gamma \log p + (1-y)p^\gamma \log(1-p)], \quad \gamma = 2.0 \quad (1)$$

- **Baseline:** $w_{\text{neg}} = 1.0$, applying equal weight to positive and negative pairs
- **Enhanced FP-Reduced:** $w_{\text{neg}} = 2.0$, explicitly amplifying the penalty on false positive matches

The motivation for the enhanced variant stems from the tracking context: a spurious Re-ID match incorrectly merges two distinct track identities, propagating an error across all subsequent frames and directly degrading MOTA. By increasing the cost of false positive predictions during training, the aim is to produce a more precision-oriented embedding model that is better suited to tracker integration. The two variants are compared on accuracy, precision, recall, F1, and raw false positive count on the validation split.

Tentative List of Tasks for Next Week:

1. Integrate the optimised Siamese Re-ID module into a custom-designed BoT-SORT tracker
2. Apply Camera Motion Compensation (CMC) via sparse optical flow to correct UAV ego-motion

3. Evaluate the full custom tracker pipeline across all 7 VisDrone2019-MOT validation sequences on GPU
4. Implement Phase 1 and Phase 2 of a structured four-phase hyperparameter sweep for further tracker improvement
5. Log all hyperparameter trials for reproducibility and rank by composite score (IDF1, HOTA, MOTA)